An Explicit $\mathbf{A}^6$ Point Producer for the Twists in Algorithm 5.1 of Elsenhans–Jahnel

Cuspidal-cubic parameters, Pinkham's E_6 action, and integration with the descent data of Section 7

June 24, 2026

Abstract

The rational-point search in step (vi) of Algorithm 5.1 can be replaced by an explicit dominant rational map from affine six-space. The construction puts six ordered points on a fixed cuspidal cubic, uses Pinkham's linear reflection representation of $W(E_6)$ on their six parameters, twists this six-dimensional representation by the same Galois cocycle used in the paper, and evaluates Coble's forty gamma coordinates. A useful normalization by the six-variable Vandermonde turns the restricted gamma coordinates into degree-nine polynomials and removes the parameter-dependent scalar in the Cremona action. This note gives all formulas, the required linear systems, the conversion to the ten descended coordinates from Section 7, and implementation-oriented pseudocode. The resulting map is dominant, so an exhaustive search in six integer parameters is guaranteed to produce a point in the open twisted marked moduli space, not merely in its compactification.

Contents

1 The conclusion and the formula to implement

Retain the notation and the data from Section 7 of the earlier note. Thus $L/\mathbf{Q}$ is finite Galois,

$$\rho : \mathrm{Gal}(L/\mathbf{Q}) \xrightarrow{\sim} G \subseteq W(E_6),$$

$J \in \mathrm{Mat}_{80\times 10}(\mathbf{Q})$ identifies the ten-dimensional span of the signed gamma coordinates with a subspace of $\mathbf{Q}^{80}$, and $B_\rho \in \mathrm{GL}_{10}(L)$ has columns forming a $\mathbf{Q}$-basis of the exact semilinear fixed space used to descend the ambient $\mathbf{P}^9$.

There is a second fixed-space matrix

$$D_\rho \in \mathrm{GL}_6(L),$$

computed from Pinkham's six-dimensional reflection representation. It identifies the twist of that representation with $\mathbf{A}^6_{\mathbf{Q}}$:

$$u \in \mathbf{Q}^6 \longmapsto t = D_\rho u \in L^6.$$

Let

$$\widehat{\Gamma} : L^6 \longrightarrow L^{80}$$

be the explicitly evaluable signed Coble vector defined in section 5. It consists of degree-nine polynomials. Choose a rational left inverse

$$K \in \mathrm{Mat}_{10\times 80}(\mathbf{Q}), \qquad KJ = I_{10}.$$

Then the desired map is

$$\boxed{\Phi_\rho : \mathbf{A}^6_{\mathbf{Q}} \dashrightarrow \widetilde{M}_\rho \subseteq \mathbf{P}^9_{\mathbf{Q}}, \qquad u \longmapsto \left[B_\rho^{-1} K \widehat{\Gamma}(D_\rho u)\right].} \tag{1}$$

Although the vector inside brackets is written with entries in L, its coordinate ratios lie in $\mathbf{Q}$. On the open set specified in section 3.2, the image lies in the open twisted marked moduli space $\widetilde{\mathcal{M}}_\rho$, hence represents a smooth marked cubic surface.

Algorithmic consequence. Steps (v) and (vi) of Algorithm 5.1—descending the thirty cubics and then searching for a rational point on their common zero locus—are no longer needed for point production. One may retain the descended cubics as an optional certificate. The replacement consists only of a six-dimensional semilinear fixed-space computation, evaluation of forty explicit degree-nine products, and linear algebra with J and B_ρ.

2 What must be added to the Section 7 data

The construction reuses all of the ten-dimensional data already described in Section 7. Only three additions are needed.

1. A six-dimensional matrix representation

$$\mu : W(E_6) \longrightarrow \mathrm{GL}_6(\mathbf{Q}), \qquad g \longmapsto M_g,$$

   with explicit generator matrices; see section 4.

2. A basis matrix D_ρ for the semilinear fixed space in L^6; see section 7.

3. An evaluator for the forty normalized gamma polynomials, in precisely the same label order used to construct J and the signed permutation matrices; see sections 5 and 14.

The safest universal data structure stores every element of $W(E_6)$ as a word in fixed generators. The same word is then evaluated in the 80-dimensional signed gamma representation, the ten-dimensional representation, and the six-dimensional Pinkham representation. This removes nearly all inverse and left-versus-right action ambiguity.

3 The untwisted cuspidal-cubic parameterization

3.1 Six points on a fixed cuspidal cubic

Let

$$C: \; yz^2 = x^3 \subseteq \mathbf{P}^2_{\mathbf{Q}}.$$

Its smooth locus is parametrized by

$$p(t) = (t : t^3 : 1), \qquad t \in \mathbf{A}^1.$$

For

$$t = (t_1, \ldots, t_6) \in \mathbf{A}^6,$$

put

$$p_i(t) = p(t_i) = (t_i : t_i^3 : 1).$$

Whenever these six ordered points are in general position, blowing them up and using the anti-canonical linear system produces a smooth cubic surface together with the ordered six exceptional curves. The latter are a marking. Thus one obtains a rational map

$$f : \mathbf{A}^6 \dashrightarrow \widetilde{\mathcal{M}},$$

where $\widetilde{\mathcal{M}}$ denotes the moduli space of smooth marked cubic surfaces.

Duncan and Reichstein explain that this map is dominant: a general ordered six-point configuration in $\mathbf{P}^2$ lies on the smooth locus of some cuspidal cubic, and all cuspidal plane cubics are projectively equivalent in characteristic different from 2 and 3 [2, Lemma 6.1].

3.2 The regular locus

For a subset $I \subseteq \{1, \ldots, 6\}$, write

$$s_I(t) = \sum_{i \in I} t_i, \qquad s_{[6]}(t) = t_1 + \cdots + t_6.$$

The six points are distinct precisely when

$$t_i - t_j \neq 0 \qquad (i \neq j).$$

Three distinct points p_i, p_j, p_k on C are collinear precisely when

$$t_i + t_j + t_k = 0,$$

and all six lie on a conic precisely when

$$t_1 + \cdots + t_6 = 0$$

[2, Lemma 6.1]. Consequently the general-position locus is

$$V^\circ = \{t \in \mathbf{A}^6 : \delta_{E_6}(t) \neq 0\},$$

where

$$\boxed{\delta_{E_6}(t) = \left(\prod_{1 \le i < j \le 6} (t_j - t_i) \right) \left(\prod_{1 \le i < j < k \le 6} (t_i + t_j + t_k) \right) (t_1 + \cdots + t_6).} \tag{2}$$

There are $15 + 20 + 1 = 36$ linear factors. They are the positive-root hyperplanes for the reflection representation of type E_6 used below.

On V°, the map f is a regular map to $\widetilde{\mathcal{M}}$. The complement of V° is exactly the locus that must be rejected before evaluating a candidate parameter vector.

4 Pinkham's explicit six-dimensional $W(E_6)$ action

4.1 Simple reflections

Regard $t = (t_1, \ldots, t_6)^t$ as a column vector. Put

$$c = t_1 + t_2 + t_3.$$

Pinkham's Cremona reflection is

$$\begin{aligned} t_i &\longmapsto t_i - \frac{2}{3}c \quad (i = 1, 2, 3), \\ t_j &\longmapsto t_j + \frac{1}{3}c \quad (j = 4, 5, 6). \end{aligned} \tag{3}$$

In matrix form,

$$M_{s_1} = \frac{1}{3} \begin{pmatrix} 1 & -2 & -2 & 0 & 0 & 0 \\ -2 & 1 & -2 & 0 & 0 & 0 \\ -2 & -2 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 3 & 0 & 0 \\ 1 & 1 & 1 & 0 & 3 & 0 \\ 1 & 1 & 1 & 0 & 0 & 3 \end{pmatrix}. \tag{4}$$

For $i = 2, \ldots, 6$, let s_i interchange t_{i-1} and t_i and leave the other coordinates fixed. These six matrices satisfy the Coxeter relations of type E_6. With the numbering used here, the edges are

$$\{1,4\},\ \{2,3\},\ \{3,4\},\ \{4,5\},\ \{5,6\}.$$

Thus

$$(s_i s_j)^3 = 1 \quad \text{for these five pairs}, \qquad (s_i s_j)^2 = 1 \quad \text{for all other } i \neq j.$$

Pinkham derives (3) from the standard quadratic transformation centered at the first three blow-up points and proves that these transformations give the Weyl reflection action [4, pp. 196–198]. Duncan–Reichstein use this observation to make the dominant map $\mathbf{A}^6 \dashrightarrow \widetilde{\mathcal{M}}$ $W(E_6)$-equivariant [2, Lemma 6.1].

4.2 The three generators matching the authors' Magma example

The supplementary file `c9_example.m` constructs $W(E_6)$ from three operations: one Cremona transformation, a six-cycle of the points, and the transposition of the first two points [5]. The corresponding matrices on t are

$$M_{\mathrm{cr}} = M_{s_1},$$

$$M_{\mathrm{cyc}} \begin{pmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \\ t_5 \\ t_6 \end{pmatrix} = \begin{pmatrix} t_2 \\ t_3 \\ t_4 \\ t_5 \\ t_6 \\ t_1 \end{pmatrix}, \qquad M_{\mathrm{sw}} \begin{pmatrix} t_1 \\ t_2 \\ t_3 \\ t_4 \\ t_5 \\ t_6 \end{pmatrix} = \begin{pmatrix} t_2 \\ t_1 \\ t_3 \\ t_4 \\ t_5 \\ t_6 \end{pmatrix}.$$

These three matrices generate the same six-dimensional representation. If the existing implementation represents elements of G as words in the three gamma matrices constructed from `bo -> bo2`, `bo -> bo3`, and `bo -> bo4`, evaluate those identical words in $M_{\mathrm{cr}}, M_{\mathrm{cyc}}, M_{\mathrm{sw}}$.

4.3 Convention test

The paper and its Magma code use row vectors in some places and column vectors in others, and a permutation of coordinates can be represented either by a matrix or its inverse. Before any descent computation, perform the following universal test for each chosen generator g and a random $t \in V^\circ(\mathbf{Q})$:

$$[\Gamma(M_g t)] = [P_g \Gamma(t)] \quad \text{in } \mathbf{P}^{79}.$$

If instead the right side agrees with $P_g^{-1}\Gamma(t)$, replace every M_g by $M_{g^{-1}}$, or equivalently reverse the word convention. The same adjustment must then be used in the six- and ten-dimensional fixed equations.

5 Coble coordinates restricted to the cusp

5.1 Factorization of the basic determinants

For $I \subseteq \{1, \ldots, 6\}$, define

$$\Delta_I(t) = \prod_{\substack{i<j \\ i,j \in I}} (t_j - t_i), \qquad \Delta_{\mathrm{VdM}}(t) = \Delta_{\{1,\ldots,6\}}(t).$$

For a three-element set $I = \{i, j, k\}$, Coble's minor evaluated at $p_r = (t_r : t_r^3 : 1)$ is

$$m_I(t) = \Delta_I(t) s_I(t). \tag{5}$$

Indeed, it is the determinant of the three rows $(t_r, t_r^3, 1)$ for $r \in I$. The conic determinant of Elsenhans–Jahnel, Definition 2.3, becomes

$$d_2(t) = \Delta_{\mathrm{VdM}}(t) s_{[6]}(t). \tag{6}$$

These identities turn every gamma coordinate into a product of linear forms.

5.2 The ten coordinates of the first type

For an unordered decomposition

$$I \sqcup I^c = \{1, \ldots, 6\}, \qquad |I| = |I^c| = 3,$$

Coble's first type is

$$\gamma_{I|I^c} = m_I m_{I^c} d_2.$$

Using (5) and (6),

$$\gamma_{I|I^c} = \Delta_{\mathrm{VdM}} \, \Delta_I \Delta_{I^c} s_I s_{I^c} s_{[6]}.$$

Define the normalized coordinate

$$\boxed{\widehat{\gamma}_{I|I^c}(t) := \frac{\gamma_{I|I^c}(t)}{\Delta_{\mathrm{VdM}}(t)} = \Delta_I(t) \Delta_{I^c}(t) s_I(t) s_{I^c}(t) s_{[6]}(t).} \tag{7}$$

It is a homogeneous polynomial of degree 9.

5.3 The thirty coordinates of the second type

Let A, B, C be three unordered pairs partitioning $\{1, \ldots, 6\}$. A cyclic orientation (A, B, C) determines one of the two coordinates attached to the partition. Coble's formula is

$$\gamma_{A \to B \to C} = \prod_{a \in A} m_{\{a\} \cup B} \prod_{b \in B} m_{\{b\} \cup C} \prod_{c \in C} m_{\{c\} \cup A}.$$

For a pair $A = \{a_1 < a_2\}$, write

$$d_A = t_{a_2} - t_{a_1}.$$

In the product above, every difference $t_j - t_i$ occurs at least once; the three within-pair differences d_A, d_B, d_C occur twice, and all twelve cross-pair differences occur once. Hence the difference part is $\Delta_{\mathrm{VdM}} d_A d_B d_C$. Therefore

$$\boxed{\begin{aligned}\widehat{\gamma}_{A\to B\to C}(t) &:= \frac{\gamma_{A\to B\to C}(t)}{\Delta_{\mathrm{VdM}}(t)} \\ &= d_A d_B d_C \prod_{a\in A} s_{\{a\}\cup B} \prod_{b\in B} s_{\{b\}\cup C} \prod_{c\in C} s_{\{c\}\cup A}.\end{aligned}} \tag{8}$$

This is again a degree-nine polynomial. Reversing the cyclic orientation from (A, B, C) to (A, C, B) gives the other coordinate.

5.4 The normalized gamma vector and the convention sign mask

The formulas (7) and (8) use the *canonical minor convention*: whenever a minor m_I occurs, its row indices are first put in increasing order. Choose exactly the same ordering of the forty *labels* as in the universal data and initially put

$$\widehat{\Gamma}^{\mathrm{can}}_{40}(t) = \big(\widehat{\gamma}_1(t), \ldots, \widehat{\gamma}_{40}(t)\big)^t.$$

Because every raw gamma coordinate has the common factor Δ_{VdM} on the cusp,

$$[\widehat{\Gamma}^{\mathrm{can}}_{40}(t)] = [\Gamma^{\mathrm{can}}_{40}(t)] \in \mathbf{P}^{39} \qquad (t \in V^\circ).$$

Thus the normalized vector gives exactly the same marked cubic surface. Its lower degree is computationally important: products of degree 9 replace products of degree 24.

There is one small but essential implementation detail. A stored implementation may evaluate a determinant with its row indices in the order appearing in Coble's product, rather than first sorting them. This changes individual gamma coordinates by fixed signs. In particular, the authors' `gamma_3` routine has this feature for some of the thirty type-II coordinates. Therefore store a universal sign mask

$$\eta = (\eta_1, \ldots, \eta_{40}) \in \{\pm 1\}^{40}$$

and define the normalized vector in the *stored* forty-coordinate convention by

$$\widehat{\Gamma}^{\mathrm{st}}_{40}(t) = \big(\eta_1\widehat{\gamma}_1(t), \ldots, \eta_{40}\widehat{\gamma}_{40}(t)\big)^t. \tag{9}$$

The mask can be computed once and for all from a single regular rational test vector $t^{(0)}$:

$$\eta_a = \frac{\Gamma^{\mathrm{st}}_a(t^{(0)})}{\Delta_{\mathrm{VdM}}(t^{(0)})\widehat{\gamma}_a(t^{(0)})} \in \{\pm 1\}. \tag{10}$$

Equivalently, for each raw product multiply the parities required to sort the row indices of its six minors. The quotient in (10) must be exactly $+1$ or -1 and should be asserted in code. The ten type-I entries in the authors' ordering have sign $+1$; among the thirty type-II entries some signs are negative.

To pass to the paper's 80 signed coordinates, use the stored signed-label list. For each signed label (ε, a), with $\varepsilon \in \{\pm 1\}$ and $1 \le a \le 40$, place $\varepsilon\eta_a\widehat{\gamma}_a(t)$ in the corresponding slot. Denote the result by

$$\widehat{\Gamma}(t) \in L^{80}.$$

Do not assume that the universal order is "all positive coordinates, then all negative coordinates" unless that is how J and P_g were constructed. The sign mask and the signed-label order are separate pieces of universal data.

6 Why the Vandermonde normalization is especially useful

The map to projective gamma coordinates is already enough for descent. The normalization above has the additional advantage that it removes the nonconstant common scalar in Pinkham's Cremona reflection.

6.1 The root-factor substitution table

Put

$$A_0 = \{1,2,3\}, \qquad B_0 = \{4,5,6\}, \qquad c = s_{A_0}.$$

Under the reflection (3):

Linear form	Image
$t_j - t_i$ with i, j both in A_0 or both in B_0	unchanged
$t_j - t_i$ with $i \in A_0$, $j \in B_0$	$s_{(A_0\setminus\{i\})\cup\{j\}}$
$s_{(A_0\setminus\{i\})\cup\{j\}}$ with $i \in A_0$, $j \in B_0$	$t_j - t_i$
s_{A_0}	$-s_{A_0}$
s_{B_0}	$s_{[6]}$
$s_{[6]}$	s_{B_0}
A triple sum with one index in A_0 and two in B_0	unchanged

Define

$$D_\times(t) = \prod_{i\in A_0}\prod_{j\in B_0}(t_j - t_i),$$

$$N_\times(t) = \prod_{i\in A_0}\prod_{j\in B_0} s_{(A_0\setminus\{i\})\cup\{j\}}, \qquad q(t) = \frac{N_\times(t)}{D_\times(t)}.$$

The table gives

$$\Delta_{\mathrm{VdM}}(M_{s_1}t) = q(t)\Delta_{\mathrm{VdM}}(t). \tag{11}$$

Substitution in Coble's forty formulas gives a signed permutation matrix Q_1 such that

$$\Gamma_{40}(M_{s_1}t) = q(t)Q_1\Gamma_{40}(t).$$

After division by Δ_{VdM}, the scalar cancels:

$$\widehat{\Gamma}_{40}(M_{s_1}t) = Q_1\widehat{\Gamma}_{40}(t). \tag{12}$$

For an adjacent transposition s_i, $i = 2, \dots, 6$, direct relabeling gives a signed permutation Q_i with

$$\Gamma_{40}(M_{s_i}t) = Q_i\Gamma_{40}(t), \qquad \Delta_{\mathrm{VdM}}(M_{s_i}t) = -\Delta_{\mathrm{VdM}}(t).$$

Consequently

$$\widehat{\Gamma}_{40}(M_{s_i}t) = (-Q_i)\widehat{\Gamma}_{40}(t). \tag{13}$$

The six signed matrices

$$\widehat{P}_{s_1} = Q_1, \qquad \widehat{P}_{s_i} = -Q_i \quad (2 \le i \le 6)$$

satisfy the E_6 Coxeter relations and yield an exact polynomial equivariance

$$\widehat{\Gamma}_{40}(M_g t) = \widehat{P}_g \widehat{\Gamma}_{40}(t) \qquad (g \in W(E_6)). \tag{14}$$

6.2 How to construct the signed matrices without interpolation

Equations (7) and (8) express each coordinate as a multiset of linear root factors. For each simple generator:

1. transform every factor using the table above or the corresponding adjacent transposition;
2. sort the transformed factors;
3. locate the unique normalized gamma label with the same factor multiset;
4. record the sign coming from factors sent to their negatives.

This produces the six 40×40 signed permutation matrices exactly. It is a symbolic alternative to the interpolation-and-matching step in the authors' Magma example. Doubling the signs in the usual way produces ordinary permutations on the 80 signed coordinates.

If $H = \mathrm{diag}(\eta_1, \dots, \eta_{40})$ is the convention sign mask, the matrices in the stored forty-coordinate convention are obtained by conjugating by H. The signed lift obtained this way and the signed lift stored in an existing implementation can still differ by harmless global signs or by the chosen ordering of the two signs, even though their projective actions agree. For the map (1), only projective agreement with the stored P_g is required. The generator test in section 4 detects the necessary conversion.

7 Twisting the six-dimensional source

7.1 The semilinear fixed space

Let $g_1, \dots, g_r$ be the same generators of G used in the ten-dimensional descent, and let

$$\sigma_i = \rho^{-1}(g_i) \in \mathrm{Gal}(L/\mathbf{Q}).$$

Suppose the convention in Section 7 imposes

$$R_{g_i} \sigma_i(y) = y \qquad \text{on } L^{10}.$$

Use the identical convention on L^6:

$$M_{g_i} \sigma_i(t) = t, \qquad i = 1, \dots, r. \tag{15}$$

The common fixed vectors form a six-dimensional $\mathbf{Q}$-vector space

$$V_\rho^{(6)} = \{t \in L^6 : M_{g_i} \sigma_i(t) = t \text{ for every } i\}.$$

This is the affine-space twist of Pinkham's reflection representation.

If the existing code instead uses the equivalent convention $\sigma_i(y) = R_{g_i} y$, then use $\sigma_i(t) = M_{g_i} t$. The point is not which formula is preferred; the point is that the six- and ten-dimensional descents must use the same descent datum.

7.2 The rational linear system

Choose a $\mathbf{Q}$-basis $\omega_1, \ldots, \omega_n$ of L, and let $S_{\sigma_i} \in \mathrm{GL}_n(\mathbf{Q})$ be the matrix of σ_i:

$$\sigma_i(\omega_b) = \sum_a (S_{\sigma_i})_{ab} \omega_a.$$

Write

$$t_j = \sum_{a=1}^{n} U_{aj} \omega_a, \qquad U \in \mathrm{Mat}_{n \times 6}(\mathbf{Q}).$$

Then (15) is equivalent to

$$S_{\sigma_i} U M_{g_i}^t = U. \tag{16}$$

After vectorization,

$$(M_{g_i} \otimes S_{\sigma_i} - I_{6n}) \operatorname{vec}(U) = 0.$$

Stack these equations for the selected generators. The common kernel must have $\mathbf{Q}$-dimension 6.

Convert a $\mathbf{Q}$-basis of this kernel to six vectors $d_1, \ldots, d_6 \in L^6$ and put

$$D_\rho = (d_1 \ \cdots \ d_6) \in \mathrm{GL}_6(L). \tag{17}$$

Then

$$t = D_\rho u, \qquad u \in \mathbf{Q}^6,$$

parametrizes the twisted source. Exact checks are

$$\det(D_\rho) \neq 0, \qquad M_{g_i} \sigma_i(D_\rho) = D_\rho \quad \text{for every generator } g_i.$$

7.3 LLL reduction of the source lattice

Coefficient growth can be reduced in the same way as in Remark 5.2 of the paper. After computing $V_\rho^{(6)}$, form a full-rank lattice such as

$$\Lambda_\rho = V_\rho^{(6)} \cap \mathcal{O}_L^6$$

after clearing denominators and saturating. Apply the Minkowski embedding and LLL to obtain a short $\mathbf{Z}$-basis. Use that basis as the columns of D_ρ. This does not alter the map; it merely replaces u by a matrix in $\mathrm{GL}_6(\mathbf{Q})$ and tends to make small integer vectors u produce much smaller number-field values.

8 Converting a source point to the Section 7 coordinates

8.1 A left inverse for J

Since J has full column rank, choose

$$K \in \mathrm{Mat}_{10\times 80}(\mathbf{Q}), \qquad KJ = I_{10}.$$

A sparse construction is preferable. Choose ten row indices $I = \{i_1, \ldots, i_{10}\}$ for which the submatrix J_I is invertible. Let $E_I : \mathbf{Q}^{80} \to \mathbf{Q}^{10}$ select those entries. Then

$$K = J_I^{-1} E_I.$$

If the ten coordinates $y_1, \ldots, y_{10}$ were chosen to be ten of the signed gamma coordinates, $J_I = I_{10}$ and K is simply the coordinate selector.

8.2 The conversion algorithm

Given $u \in \mathbf{Q}^6$, execute the following steps.

P1. Compute

$$t = D_\rho u \in L^6.$$

P2. Test the 36 factors in (2). If any factor vanishes, reject u.

P3. Evaluate the forty degree-nine polynomials (7)–(8), insert signs in the stored 80-coordinate order, and obtain

$$x = \widehat{\Gamma}(t) \in L^{80}.$$

P4. Recover the original ten coordinates

$$y = Kx \in L^{10}.$$

As an optional check, verify $Jy = x$.

P5. Convert to the descended basis:

$$w = B_\rho^{-1} y \in L^{10}.$$

P6. Choose an index j with $w_j \neq 0$ and set

$$z_i = \frac{w_i}{w_j}, \qquad i = 1, \ldots, 10. \tag{18}$$

Every z_i lies in $\mathbf{Q}$. Coerce them exactly to rational numbers and return

$$P = [z_1 : \cdots : z_{10}] \in \widetilde{M}_\rho(\mathbf{Q}).$$

P7. For the subsequent Clebsch calculation, use the exact fixed representative

$$y_0 = B_\rho z = \frac{y}{w_j}, \qquad x_0 = Jy_0 = \frac{x}{w_j}.$$

Extract the designated forty positive gamma entries from x_0. Their power-sum expressions give rational Clebsch invariants exactly as in the original algorithm.

8.3 Why the ratios are rational

Let $\sigma \in \mathrm{Gal}(L/\mathbf{Q})$ and $g = \rho(\sigma)$. Since $t = D_\rho u$ is fixed by the twisted source action and the cusp map is $W(E_6)$-equivariant, the gamma point satisfies the projective descent relation. With the convention used here, there is a scalar $c_\sigma \in L^\times$ such that

$$P_g \sigma(x) = c_\sigma x.$$

Because $x = Jy$ and $P_g J = J R_g$, injectivity of J gives

$$R_g \sigma(y) = c_\sigma y.$$

Write $y = B_\rho w$. The columns of B_ρ are exact fixed vectors, so

$$R_g \sigma(B_\rho) = B_\rho.$$

It follows that

$$\sigma(w) = c_\sigma w.$$

Therefore

$$\sigma\left(\frac{w_i}{w_j}\right) = \frac{w_i}{w_j}$$

for every σ, proving $w_i/w_j \in \mathbf{Q}$. This argument is why the raw gamma vector need only satisfy projective, rather than vector, equivariance.

9 The modified version of Algorithm 5.1

9.1 Universal precomputation

Compute once:

1. the forty gamma labels, the convention sign mask η, and the 80 signed-label order;
2. J, the ten-dimensional matrices R_g, and optionally the thirty universal cubic relations, as in the paper;
3. the six Pinkham generator matrices and a word map that evaluates the same abstract $W(E_6)$ element in dimensions 80, 10, and 6;
4. an evaluator for (7) and (8) that applies the stored sign mask (9);
5. a left inverse K of J.

9.2 Instance-specific computation

Given (L, G, ρ):

Step 1. Compute the same generator/automorphism pairs (g_i, σ_i) as in Algorithm 5.1.

Step 2. Compute B_ρ from the ten-dimensional semilinear fixed equations, exactly as in Section 7.

Step 3. Evaluate the words for g_i in the six-dimensional representation to obtain M_{g_i}, and solve (16) to obtain D_ρ.

Step 4. Enumerate primitive vectors $u \in \mathbf{Z}^6$ by increasing height. For each one, apply steps P1–P6 of section 8. Reject u if it lies on a root hyperplane.

Step 5. Optionally verify that the returned rational point z satisfies the thirty descended cubics. These cubics are no longer needed to find z.

Step 6. Use $x_0 = JB_\rho z$ to compute the forty exact gamma values, the Clebsch vector $[A : B : C : D : E]$, the pentahedral quintic, and the trace cubic of Algorithm A.10.

Step 7. If $E = 0$, the pentahedral quintic is inseparable, or the resulting cubic is singular, continue with the next vector u.

9.3 Language-neutral pseudocode

```
function PointFromA6(instance_data, universal_data):
    # Existing Section 7 objects:
    # B_rho, J, K, gamma_sign_mask, signed_gamma_order,
    # Clebsch formulas
    # New object:
    # D_rho, obtained from the 6-dimensional fixed equations

    for u in primitive_integer_vectors_by_height(6):
        t = D_rho * u # vector in L^6

        if any_E6_root_factor_is_zero(t):
            continue

        gamma40_can = canonical_normalized_cusp_gammas(t)
        gamma40 = apply_sign_mask(gamma40_can, gamma_sign_mask)
        x = insert_signed_coordinates(gamma40)
        y = K * x

        assert J * y == x

        w = inverse(B_rho) * y
        j = first_nonzero_coordinate(w)
        z = [w[i] / w[j] for i in 1..10]

        if not all_entries_are_exactly_rational(z):
            error("action convention or coordinate ordering mismatch")

        z = coerce_to_Q(z)
        x_exact = J * B_rho * z

        # Optional certificate:
        assert all(descended_cubic(z) == 0)

        clebsch = clebsch_from_positive_gammas(x_exact)
        if clebsch.E == 0:
            continue
```

```
    surface = algorithm_A10(clebsch)
    if surface_failed or surface_is_singular(surface):
        continue

    return z, surface
```

10 Direct integration with `c9_example.m`

The authors' example already contains nearly all universal objects:

- `all_gamma(mat)` evaluates the forty raw gamma coordinates;
- `bm` is a 10×40 row-basis matrix for their span;
- `se` gives the forty linear forms in ten coordinates;
- `mat_l` contains the three 40×40 gamma matrices for the Cremona generator, the six-cycle, and the transposition;
- `z_bm_o_10` and `para` encode the descended ten-dimensional basis.

A minimally invasive patch has the following structure.

1. Store the three source matrices $M_{\mathrm{cr}}, M_{\mathrm{cyc}}, M_{\mathrm{sw}}$ from section 4.
2. Whenever an element of the subgroup is constructed, retain a word in the three generators. Evaluate the word both in `mat_l` and in the three source matrices. This gives the matching 40- and 6-dimensional matrices without a group-identification problem.
3. Construct the six-dimensional semilinear fixed space in exact analogy with the block-matrix construction used for the ten-dimensional lattice. Its rank is six. Apply LLL if desired and convert the result to D_ρ.
4. Delete the nested ten-variable point search. Enumerate six integers, form $t = D_\rho u$, and build the 6×3 matrix with rows
$$(t_i, t_i^3, 1).$$
The existing `all_gamma` function may be used directly. Divide its output by
$$\prod_{i<j}(t_j - t_i)$$
to obtain normalized values in exactly the authors' sign convention. After this version passes all checks, one may replace it by the degree-nine product formulas, provided the universal sign mask (9) is applied.
5. Solve for the original ten coordinates using `bm`, then solve for the descended coordinates using the matrix encoded by `para`. Normalize one nonzero descended coordinate to 1 and verify that the remaining ratios are rational.

6. Map the rational descended vector forward through `para` and `se`. Use this exact fixed gamma representative in the existing power-sum and Clebsch code.

Row-vector warning. The paper's example uses expressions of the form `row * bm` and therefore transposes several matrices relative to the column convention in this note. Translate the equations structurally, not by copying the transpose symbols. The two decisive tests are that the six-dimensional fixed space has dimension 6 and that the final descended ratios are fixed by every field automorphism.

---

11 Existence, dominance, and termination of the search

11.1 Why every twist receives such a map

The untwisted map

$$f : \mathbf{A}^6 \dashrightarrow \widetilde{\mathcal{M}}$$

is dominant and $W(E_6)$-equivariant [2, Lemma 6.1]. Twisting both sides by the cocycle ρ gives a dominant rational map

$$f_\rho : {}^\rho\mathbf{A}^6 \dashrightarrow \widetilde{\mathcal{M}}_\rho.$$

The twist of a linear representation is a six-dimensional $\mathbf{Q}$-vector space; the matrix D_ρ explicitly identifies it with $\mathbf{A}^6_{\mathbf{Q}}$. Formula (1) is the Coble-coordinate realization of this twisted map. Dominance may be checked after base change to L, where it becomes the original map composed with the invertible linear change of variables D_ρ.

It follows that

$$\widetilde{\mathcal{M}}_\rho(\mathbf{Q})$$

is Zariski dense. In particular, the point sought in Algorithm 5.1 always exists. The earlier bounded search could fail only because its chosen height bound was too small.

11.2 Why exhaustive integer enumeration terminates

The pullback of the boundary is contained in the zero set of

$$\delta_{E_6}(D_\rho u) \in L[u_1, \ldots, u_6],$$

a nonzero polynomial. Hence rational vectors avoiding the boundary are Zariski dense in $\mathbf{A}^6_{\mathbf{Q}}$.

The conditions $E = 0$, $\Delta = 0$, and $F = 0$ used in the paper define proper divisors in the target. Since Φ_ρ is dominant, their pullbacks are proper closed subsets of $\mathbf{A}^6$. Therefore an enumeration of primitive integer vectors that is exhaustive by height must eventually find a vector avoiding the root hyperplanes and all three downstream failure loci. No useful small universal height bound follows from this argument, but the search is no longer heuristic in principle.

12 Certification and debugging checklist

A robust exact implementation should perform the following checks.

12.1 Universal checks

1. Verify the Coxeter relations for the six matrices M_{s_i}.
2. For random $t \in V^\circ(\mathbf{Q})$ and every universal generator, verify
$$[\widehat{\Gamma}(M_g t)] = [P_g \widehat{\Gamma}(t)].$$
3. Verify the canonical degree-nine formulas against determinant evaluation:
$$\Gamma_{40}^{\mathrm{can}}(t) = \Delta_{\mathrm{VdM}}(t)\widehat{\Gamma}_{40}^{\mathrm{can}}(t).$$
Then verify the stored evaluator using the sign mask:
$$\Gamma_{40}^{\mathrm{st}}(t) = \Delta_{\mathrm{VdM}}(t)\widehat{\Gamma}_{40}^{\mathrm{st}}(t).$$
4. Verify $KJ = I_{10}$ and $P_g J = JR_g$.
5. If constructing the normalized signed action symbolically, verify all Coxeter relations for the resulting signed permutation matrices.

12.2 Instance checks

1. The ten-dimensional fixed space has dimension 10 and $\det(B_\rho) \neq 0$.
2. The six-dimensional fixed space has dimension 6 and $\det(D_\rho) \neq 0$.
3. For every generator,
$$M_{g_i}\sigma_i(D_\rho) = D_\rho, \qquad R_{g_i}\sigma_i(B_\rho) = B_\rho,$$
with the chosen convention.
4. For a candidate u, verify all 36 root factors are nonzero.
5. Verify $Jy = x$ after $y = Kx$.
6. Verify all ratios in (18) lie in $\mathbf{Q}$ exactly. Failure here almost always indicates an inverse, transpose, gamma-order, convention-sign-mask, or signed-label mismatch.
7. Verify $x_0 = JB_\rho z$ is exactly fixed under the 80-coordinate descent.
8. If the thirty descended cubics were computed, verify that all vanish at z.
9. Verify the Clebsch quantities computed from x_0 are rational and that the final cubic is smooth.
10. For a final independent check, compute the 27 lines and their Galois action.

13 Practical refinements

13.1 Evaluate products, not determinants

The formulas (7) and (8) need only additions, subtractions, and products in L. They avoid 6×6 determinants and reduce the total degree from 24 to 9. Precompute the following arrays for each t:
$$d_{ij} = t_j - t_i \quad (i < j),$$

$$s_{ijk} = t_i + t_j + t_k \quad (i < j < k), \qquad s_{[6]} = t_1 + \cdots + t_6.$$

Every normalized gamma coordinate is then a product of nine entries from these arrays.

13.2 Search order

After LLL reduction of the source lattice, enumerate primitive vectors by increasing

$$\|u\|_\infty = \max_i |u_i|$$

or by increasing Euclidean norm. Test the 36 root factors before any gamma evaluation. Since the map is homogeneous of degree 9 in t, scalar multiples of u give the same projective gamma point; primitive vectors avoid redundant work.

13.3 Do not invert B_ρ repeatedly

Compute B_ρ^{-1} once. Better still, solve the linear system $B_\rho w = y$ for each candidate using a stored decomposition. The same applies to K if it was not chosen as a selector.

13.4 Use the exact fixed representative downstream

The raw source evaluation $x = \widehat{\Gamma}(D_\rho u)$ is generally only projectively fixed relative to the particular signed lift P_g used in Section 7. Do not coerce its power sums individually to $\mathbf{Q}$. First compute z from the coordinate ratios, then replace x by

$$x_0 = J B_\rho z.$$

This vector is exactly fixed and is the correct input to the Clebsch formulas.

14 Canonical enumeration of the forty labels

A convenient abstract ordering is as follows.

14.1 Ten type-I labels

Enumerate all three-element subsets $I \subseteq \{1, \ldots, 6\}$ containing 1. For each, use the unordered label $I \mid I^c$. This gives ten labels.

14.2 Thirty type-II labels

Enumerate the fifteen perfect matchings of $\{1, \ldots, 6\}$. Sort the three pairs lexicographically as $A < B < C$. For each matching include the two cyclic orientations

$$A \to B \to C, \qquad A \to C \to B.$$

This gives thirty labels.

14.3 The ordering in the authors' example

The function `all_gamma` in `c9_example.m` places the thirty type-II coordinates first and the ten type-I coordinates second. In one-based pseudocode, its type-II part is:

```
for i,j,k,l,m in {2,...,6} with all five distinct:
    if j < k and l < m:
        A = {1,i}; B = {j,k}; C = {l,m}
        append gamma(A -> B -> C)
```

Its type-I part is:

```
for i,j,k,l,m in {2,...,6} with all five distinct:
    if i < j and k < l and l < m:
        I = {1,i,j}
        Ic = {k,l,m}
        append gamma(I | Ic)
```

These loops reproduce the authors' *coordinate order*. To reproduce their coordinate signs as well, apply the mask (9). It can be calibrated by (10), or computed directly as follows: for each entry returned by `gamma_3`, take the product of the signs of the permutations that sort the six row-index triples used by that routine. Thus no coordinate permutation is required, but the sign calibration must not be omitted. The existing `all_gamma` routine automatically supplies the correct signs; dividing its output by Δ_{VdM} is therefore the safest first implementation.

15 A compact data-flow diagram

The complete replacement for the nonlinear point search is

$$\boxed{\begin{array}{c}
(g_i,\sigma_i),\ M_{g_i} \\
\Downarrow\ \text{six-dimensional semilinear fixed equations} \\
D_\rho \in \mathrm{GL}_6(L) \\
\Downarrow\ u \in \mathbf{Z}^6 \mapsto t = D_\rho u \\
\widehat{\Gamma}(t) \in L^{80} \\
\Downarrow\ K \\
y \in L^{10} \\
\Downarrow\ B_\rho^{-1} \text{ and projective normalization} \\
z \in \mathbf{Q}^{10}, \quad [z] \in \widetilde{\mathcal{M}}_\rho(\mathbf{Q}) \\
\Downarrow\ JB_\rho z \\
\text{exact gamma values} \to [A:B:C:D:E] \to \text{Algorithm A.10.}
\end{array}}$$

16 Final assessment

The map can be made explicit enough for direct implementation. The essential new facts are:

1. Pinkham's reflection formula gives rational 6×6 matrices for the same $W(E_6)$ elements already used in Algorithm 5.1.

2. On the cuspidal cubic, every Coble gamma coordinate has a common Vandermonde factor.

Dividing it out gives the explicit degree-nine formulas (7) and (8).

3. Twisting the six-dimensional source is the same kind of rational linear algebra already used for the ten-dimensional ambient space.

4. Formula (1) converts any regular source parameter into a rational point of the descended compactification; in fact it lands in the open marked moduli space.

5. Dominance guarantees that exhaustive six-variable enumeration eventually avoids every boundary and reconstruction failure divisor.

The only delicate implementation issue is convention alignment among the abstract Weyl-group word, its action on the six parameters, and its action on the signed gamma coordinates. The generator-level projective equivariance test and the rational-ratio test isolate this issue immediately.

References

[1] A.-S. Elsenhans and J. Jahnel, *Moduli spaces and the inverse Galois problem for cubic surfaces*, Trans. Amer. Math. Soc. **367** (2015), no. 11, 7837–7861; arXiv:1209.5591.

[2] A. Duncan and Z. Reichstein, *Pseudo-reflection groups and essential dimension*, arXiv:1307.5724, especially Lemma 6.1.

[3] A. Duncan and Z. Reichstein, *Versality of algebraic group actions and rational points on twisted varieties*, J. Algebraic Geom. **24** (2015), 499–530; arXiv:1109.6093.

[4] H. Pinkham, *Résolution simultanée de points doubles rationnels*, in *Séminaire sur les singularités des surfaces*, Lecture Notes in Math. **777**, Springer, 1980, 179–203; see especially pp. 196–198.

[5] J. Jahnel, *c9_example.m: Magma example for Algorithm 5.1*, authors' supplementary webpage, `https://wwwuser.gwdguser.de/~jjahnel/Arbeiten/Kub_Fl/c9_example.m`.